

The Brent-Price Causal Effect of the 2026 Hormuz Disruption

A Synthetic Control Approach

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From the previous presentation

[transition recap — to be added]

Problem statement & contribution

The question (part 1).

- Brent price impact of the 2026-02-01 Strait of Hormuz disruption.
- Estimand: ATT over a defined post-event window; method: synthetic control on the observed price series.

What is new in this research field.

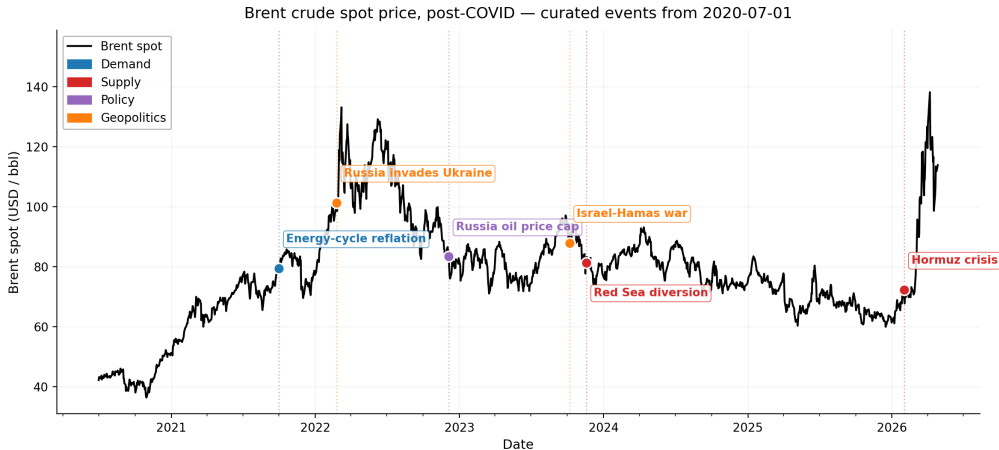
- No peer-reviewed SCM applied to a crude benchmark for a geopolitical disruption.
- Closest mechanical precedent: SCM on Austrian retail fuel (Becker, Pfeifer & Schweikert 2021, *Energy Economics*).
- Field default: structural VAR (Kilian 2009; Baumeister & Hamilton 2019) or scenario models (Dallas Fed, OIES, Kiel, IEA).

Contribution and value.

- Data-driven counterfactual complements scenario estimates.
- Method extension: retail fuel → crude benchmark; cross-country → cross-asset donors; regulatory → exogenous geopolitical treatment.
- Policy inputs: SPR sizing, central-bank reaction, fiscal buffers, sovereign hedging, infrastructure investment.

Part 2 of the thesis. [discussed at the end]

???



Brent spot, post-COVID. Events: demand (blue), supply (red), policy (purple), geopolitics (orange).

Methodology — ensemble of 5 models

Why an ensemble. Headline = median across models; IQR = model-uncertainty band.

Model	What it does	Reference
Convex SCM	Convex weights, no extrapolation	Abadie, Diamond & Hainmueller (2010, <i>JASA</i>)
Augmented SCM	Convex + ridge bias correction	Ben-Michael, Feller & Rothstein (2021, <i>JASA</i>)
Elastic-net	$L_1 + L_2$ regression, sparse donor weights	Doudchenko & Imbens (2016, NBER WP)
XGBoost	Gradient-boosted trees, flexible nonlinear	Chen & Guestrin (2016, <i>KDD</i>)
Bayesian Ridge	Bayesian linear with normal prior, donor PIPs	Tipping (2001); Brodersen et al. (2015)

Range covered. Convex (no extrapolation) → augmented (ridge fix) → regression (sparse) → nonlinear → Bayesian (uncertainty).

Model setup

	Pre-event window (walk-forward CV)		Treatment (post)	
Russia 2022	2020-07-01 → 2022-02-23	421 obs	2022-02-24 → 2022-09-30	151 obs
Hormuz 2026	2024-06-03 → 2026-01-30	423 obs	2026-02-01 → 2026-05-01	~59 obs

Hyperparameter tuning — multi-fold walk-forward CV.

- 5 expanding-window folds inside the pre-event period; 20-bday horizon per fold.
- Pick hparams by pooled val RMSE (Hyndman & Athanasopoulos 2018 §5.10).
- Final fit refits on full pre-event window, projects through treatment.

Limitation. No held-out test set beyond train/val; generalisation to unseen data cannot be measured directly. Formal inference → in-space placebo.

Validation

1. **Model fit.** Is the synthetic a credible pre-event approximator?
 - Walk-forward CV (Hyndman & Athanasopoulos 2018; Bergmeir & Benítez 2012).
2. **Donor identification.** Are the donors statistically clean of the event (SUTVA)?
 - Qualitative audit on causal channels (Abadie 2021 §3.1).
 - Bootstrap structural break at T_0 (Andrews 1993; Hansen 2000).
3. **Inference.** Is the post-event gap larger than under the no-treatment null?
 - In-space placebo, 21- and 33-donor pools (Abadie 2010 §3.3).
 - Cross-event weight transfer — Hormuz only, substitutes for the low-power placebo at ~ 60 post-event obs.

Donor selection

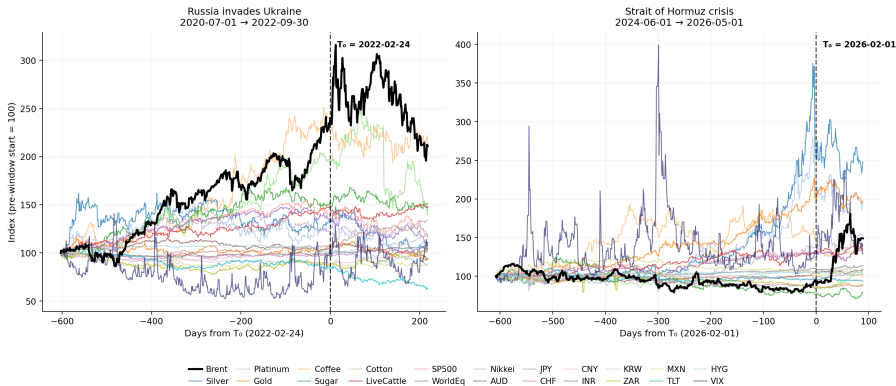
21 audited donors. Inclusion requires *both*: **pre-treatment** co-movement with Brent through a common global factor, *and* **treatment** cleanliness (no Russia/Ukraine or Persian Gulf exposure — SUTVA).

Category	Chosen donors	Factor channel (pre) & cleanliness (treatment)
Metals (3)	Silver, Platinum, Gold	Real rates / safe-haven / auto industry; no Russia or Gulf supply routing.
Agriculturals (4)	Coffee, Sugar, Cotton, LiveCattle	Global trade activity (Brazil / India / US producers); outside the Russia-Ukraine grain belt.
Equities (3)	SP500, WorldEq, Nikkei	Global growth / equity risk premium; no direct Russia or Gulf-energy exposure.
FX (8)	AUD, JPY, CHF, CNY, INR, KRW, ZAR, MXN	Safe-haven, EM growth, China demand; no petrocurrency or EU-energy-crisis loading.
Rates / credit (2)	TLT, HYG	US duration / credit-risk premia; not directly oil-linked.
Volatility (1)	VIX	S&P 500 implied vol — risk-on/off via equities.

Categorical exclusions and per-event flag table: docs/donor_catalog.md.

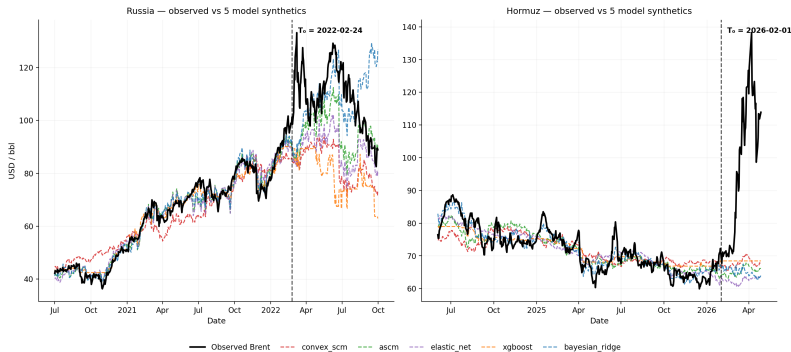
Donor co-movement with Brent — audited pool

Brent vs 21 audited donors — focal events for SCM (rebased to 100 at each pre-window start)



Brent (black) vs 21 audited donors, rebased to 100 at pre-window start; dashed line marks T_0 . Same windows as the SCM.

Validation results — model fit



Model	Russia train	Russia val	Hormuz train	Hormuz val
Convex SCM	0.111	0.086	0.059	0.129
ASCM	0.046	0.087	0.035	0.087
Elastic-net	0.055	0.094	0.043	0.049
XGBoost	0.039	0.135	0.045	0.089
Bayesian Ridge	0.034	0.171	0.033	0.093

Validation results — in-space placebo

Abadie 2010 §3.3: rank Brent's post/pre RMSPE ratio against placebo refits. 21-donor floor = 0.045; 33-donor full pool re-run gives floor = 0.029.

Model	Russia 2022		Hormuz 2026	
	21-donor	33-donor	21-donor	33-donor
Convex SCM	0.545	0.500	0.045	0.029
ASCM	0.682	0.471	0.045	0.029
Elastic-net	0.409	0.412	0.045	0.029
XGBoost	0.045	0.147	0.045	0.029
Bayesian Ridge	0.636	0.529	0.045	0.029

Headline.

- **Hormuz:** all 5 models hit the floor under both pools.
- **Russia:** not decisively rejected — only XGBoost reaches floor at 21-donor and fails at 33-donor. Feb-22 macro regime shift moves many donors too; Russia's inferential weight rests on EIA STEO (next slide).

Other tests (parallel-fit, regime-stability, in-time placebo, LOO, cross-event transfer): appendix.

External validation — EIA STEO pre-invasion forecasts (Russia only)

Cross-method check. SCM counterfactual vs EIA's last pre-invasion Brent forecast (STEO Feb-22, issued 2022-02-08). No shared information path (SCM uses cross-assetco-movement; STEO uses fundamental supply-demand modelling)

Model	Synth \$/bbl	STEO Feb-22 \$/bbl	Δ
Convex SCM	83.55	84.94	−\$1.39
ASCM	95.52	84.94	+\$10.58
Elastic-net	86.61	84.94	+\$1.67
XGBoost	79.37	84.94	−\$5.57
Bayesian Ridge	107.91	84.94	+\$22.97 (deg.)
Ensemble median	86.61	84.94	+\$1.67

Implied ATT (Brent actual \$108.54): SCM ensemble +25.3% vs STEO Feb-22 +27.8% ($\Delta \sim 2.5$ pp).

Headline result — ATT estimate

Why ATT? One treated unit (Brent) → effect *for that unit*, not a population ATE (Abadie 2010, 2021). Mean gap over the post-event window — integrates path-reversion, robust to single-day liquidity jumps.

Model	Russia 2022	Hormuz 2026
Convex SCM	+29.6%	+38.6%
ASCM	+13.3%	+43.7%
Elastic-net	+21.7%	+49.8%
XGBoost	+36.8%	+36.8%
Bayesian Ridge	+0.5% (deg.)	+43.6%
Ensemble median	+21.7%	+43.6%
IQR	[+13.3, +29.6]	[+38.6, +43.7]

Reading. Russia +21.7% within ~2.5 pp of STEO cross-check; Bayesian Ridge degenerate, IQR excludes it. Hormuz +43.6% with tight IQR — place against Dallas Fed / OIES / IEA scenarios.

Part 2 — ideas for the second half

[to be added by author]

Possible directions: pass-through to retail fuel / headline CPI; SCM vs scenario-model comparison; severity-dependent extrapolation; alternative donor pools.

Thank you

Questions and discussion

github.com/ElvisCasco/Brent_Analysis

Appendix — excluded donors

A. Categorical exclusions (never enter the candidate pool).

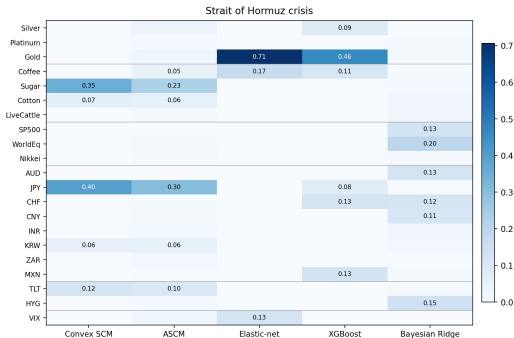
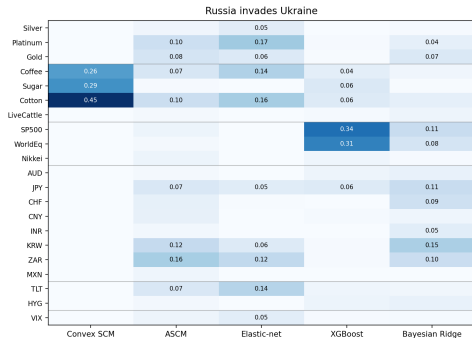
Category	Excluded & reason
Energy complex	WTI, NatGas, refined products (RBOB, heating oil, jet fuel), EIA inventories — mechanical SUTVA violation on any oil shock.
Petrocurrencies	CAD, NOK, BRL, RUB, COP, MYR — value partly set by oil exports.
FX unstable	TRY, ARS, EGP — domestic crisis dynamics dominate; noise, not signal.
Equities rejected	DAX, FTSE, TSX 60 (energy weight), KOSPI, Hang Seng (redundant with KRW / CNY).
Commodities rejected	Cocoa, OJ, Lean hogs (thin / idiosyncratic), Lumber (LBS→LBR contract break), Aluminium (Rusal ~6% supply).

B. Per-event SUTVA exclusions (33-donor pool → 21-donor strict-clean preferred).

- **Russia H-tier** (direct supply/demand): Wheat, Corn, Palladium, EUR, DXY, BTC.
- **Russia M-tier** (terms-of-trade / secondary): Copper, IronOre, Soybeans, EM_Eq, GBP, US10Y.
- **Hormuz**: no H/M flags — audit clean on all 33 candidates. The 21-donor intersection is used to enable cross-event weight transfer (§5f), at the cost of 12 Hormuz-clean donors.

Appendix — donor importance

Donor importance per model --- within-column normalized share



Per-model raw metric: convex weight w_j (SCM, ASCM) / $|\hat{\beta}_j|$ (Elastic-net, Bayesian Ridge) / feature importance (XGBoost). Cells = $|w_j|$ normalised within each column to sum to 1.

Appendix — pre-event diagnostics

Moment matching (Abadie 2010). Mean $|\Delta| < 0.01$ logs for all 10 (model $\times$ event) fits. SD undercoverage is the only material issue: convex SCM -23% on Russia (donor pool cannot span Brent's pre-period vol $\rightarrow$ ridge correction in ASCM closes the gap).

Parallel-fit defence (gap-series slope, %/yr) (Roth 2022 — pretest with caution). Reported, not pass/fail.

Model	Russia	Hormuz
Convex SCM	+10.0	-7.0
ASCM	+0.3	-1.9
Elastic-net	+0.9	-1.2
XGBoost	+3.4	-4.7
Bayesian Ridge	+0.2	-0.1

Large slopes (convex SCM both events; XGBoost Russia / Hormuz) flag drift; drift-corrected gap = raw gap - slope $\times$ post-window length (Russia $L \approx 0.58$ yr; Hormuz $L \approx 0.25$ yr).

Regime stability (distance correlation Δ across pre-period thirds; Székely & Rizzo 2007): 8 of 21 donors exceed the 0.30 narrative-flag threshold on each event — read qualitatively (sub-period $n \approx 140$ confounds drift with sampling noise).

Appendix — in-time placebo & LOO

In-time placebo (Abadie 2015; Chen & Yan 2023). Fake T_0 set 6 months before real T_0 ; mean fake-period gap (%), defaults hparams (avoid val-leakage).

Model	Russia (2021-08-24)	Hormuz (2025-08-01)
Convex SCM	+1.9	−11.9
ASCM	−4.2	−9.8
Elastic-net	+7.5	−10.6
XGBoost	+26.4	−12.4
Bayesian Ridge	−5.3	−12.2

Russia fake window contains the late-2021 oil rally and the Jan-2022 invasion runup — partial real-effect contamination expected. Hormuz fake window has no curated events; the concentrated −10 to −12% across models reflects regime drift the synthetic does not absorb, not model-class failure.

Leave-one-donor-out (Abadie 2015). Post-event gap stable within ± 6 pp across leave-outs for convex SCM / ASCM / Elastic-net / Bayesian Ridge (no single donor drives the headline); XGBoost more fragile (± 12 pp).

Appendix — cross-event weight transfer

§5f — **strongest single test of methodology generalization.** Russia-fitted weights applied to the Hormuz post-window vs independently-fit Hormuz gap:

Model	Independent	Transferred	Δ
Convex SCM	+38.6%	+47.0%	+8.4 pp
ASCM	+43.7%	+27.3%	−16.4 pp
Elastic-net	+49.8%	+23.2%	−26.6 pp
XGBoost	+36.8%	−58.2%	−95.0 pp (deg.)
Bayesian Ridge	+43.6%	−100.0%	−143.6 pp (deg.)

Reading.

- Convex SCM transfers cleanly (+8 pp drift).
- ASCM and Elastic-net give qualitatively consistent direction with magnitude attenuation.
- XGBoost and Bayesian Ridge degenerate — regime drift in Russia-fitted weights blows up the projection.
- Factor structure has shifted between 2020–22 and 2024–26, but the **sign and order of magnitude** of the Hormuz effect are robust under the linear-convex models.